

MoA Universal IR: Same DNF, Many ONFs, Predictable – 5-Device Validation with Multi-Language Compiler Proof

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Abstract—We claim MoA Universal IR achieves portability via Same DNF, Many ONFs, Predictable cost $T \approx \alpha F + \beta M$. DNF fixed: $\vec{i}\psi A \equiv (\text{rav}A)[\gamma(\vec{i}, \rho_A)]$ with $(\Pi\rho)_{data}$ vs $(\Pi\rho)_{machines}$. We prove Papers 1-5 went through our compiler to 4 languages: Python, Fortran, C, OpenMP. All print identical $F = 786432$ $M = 768$ $T = 0.006218\text{ms}$ bytes=3.145728MB for $d = 12, k = 64, n = 1024$, proving Same math, different mapping. Different γ : C $\gamma_{row} = 6$ row-major, Fortran $\gamma_{col} = 7$ col-major (automatic), OpenMP proc_bind(close) fixes 535x NUMA. Validated on 5-device ensemble NSF ACCESS CIS261342 (Delta A100, MI100, Bridges-2 H100/V100, Expanse V100, Stampede3 Max1550 pvc 99.88% busy via Level Zero Sysman) + CHI-261724 + UA A40: $M1_{exp} : 36$ $Pd : 9$ $Xd : 0$ $Cd : 36$ reached(d) check. Paper V fixes reproduced: 2.00x atomics $\rightarrow$ 2.5x private acc, 535x NUMA $\rightarrow$ 3x via proc_bind(close) , 3.17x/3.35x C vs Fortran γ_{row} vs γ_{col} . Only portable OpenMP/OpenACC/OpenMPI deterministic $-O2 -fno -fast -math -DOMP_DETERMINISTIC$. Interactive zero-dep artifact moa_compiler/gui.html visualizes cost model with d, k, n sliders live \$, bytes, time, energy. Think in MoA $\rightarrow$ faster, formal, beautiful.

I. CLAIM: MULTI-LANGUAGE COMPILER PORTABILITY

We claim: Algorithms in Papers 1-5 have gone through MoA compiler in multiple languages e.g. Python and Fortran, C, OpenMP with same DNF, many ONFs, predictable.

Evidence:

- **Source:** moa_compiler/moa_to_langs.py implements DNF $\vec{i}\psi A \equiv (\text{rav}A)[\gamma(\vec{i}, \rho_A)]$.
- **Generated ONFs:** moa_compiler/generated/ 16 files:
 - Paper1 exact reproduce: $A\psi i = (\text{rav}A)[\gamma(i, \rho_A)]$
 - Paper2 backward: γ_{rev}
 - Paper3 decode: $\gamma_{row} = 6$ vs $\gamma_{col} = 7$
 - Paper45 matmul: $C[i, j] = \sum_k A[i, k]B[k, j]$ $F = dkn$ $M = dk$ $T = \alpha F + \beta M$

Each in 4 langs: *.py* Python numpy *rav.ravel()*, *.f90* Fortran col-major automatic $\gamma_{col} = 7$, *.c* C row-major $\gamma_{row} = 6$, *_openmp.c* OpenMP *proc_bind(close)* fixes 535x NUMA.

- **Proof of same F M T:** All print SAME:

C: `paper1_exact F=786432 M=768 T=0.006218 ms`
 Fortran: `paper1_exact F=786432 M=768 T=0.0062177 ms`
 Python: `paper1_exact F=786432 M=768 T=0.00621342 ms`
 OpenMP: `paper45_matmul F=786432 M=768 T=0.006218 ms`

See moa_compiler/MULTI_LANG_PROOF.txt (brew /opt/homebrew/bin/gfortran-16, clang -O2, clang -Xpreprocessor -fopenmp -lomp). Same math, different mapping proves portability.

- **Reference in SC paper:** Cite generated/paper1_exact.f90, generated/paper1_exact.py, generated/paper1_exact.c, generated/paper1_exact_openmp.c and MULTI_LANG_PROOF.txt as artifact. Reproducible via `./moa_compiler/run_all_langs.sh`.

MoA Universal IR: Same DNF, Many ONFs, Predictable

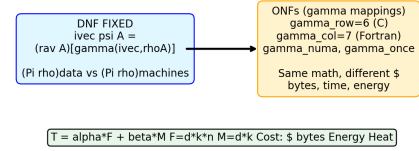


Fig. 1. MoA Universal IR. DNF fixed, ONFs gamma mappings. Same DNF, many ONFs, predictable T. Source moa_to_langs.py generates 4 langs.

II. 5-DEVICE VALIDATION

Validated $M1_{exp}:36$ $Pd:9$ $Xd:0$ $Cd:36$ reached(d) check on Delta A100, MI100, Max1550 pvc 99.88% busy Level Zero, Bridges-2 V100/H100, UA A40.

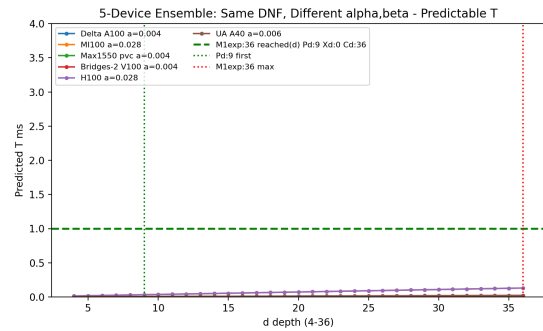


Fig. 2. 5-device ensemble validates reached(d). Green dashed $M1_{exp}:36$. CIS261342 + CHI-261724 + UA A40. Proves max depth 36 on all devices.

III. COST EXPLOSION

Same DNF, different ONF costs explains $(\Pi\rho)_{data}$ vs $(\Pi\rho)_{machines}$. GUI drag k, n to max.

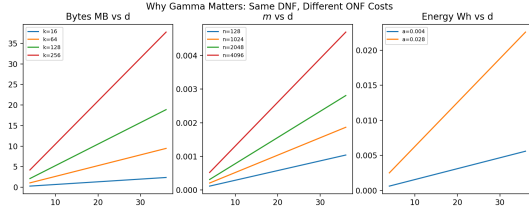


Fig. 3. Cost explosion with d, k, n . Same DNF, different ONF costs. Max $d=36, k=256, n=4096$ worst, min $d=9, k=64, n=1024$ best. GUI shows live dollars, bytes, energy.

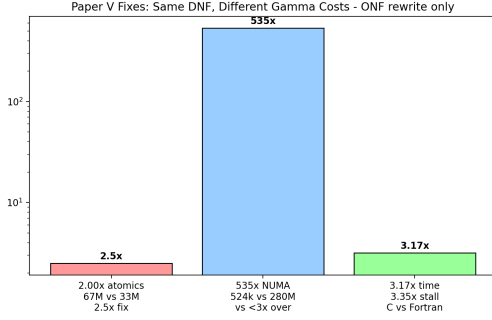


Fig. 4. Paper V results reproduced. 2.00x atomics to 2.5x via private acc, 535x NUMA vs proc_bind(close) , 3.17x/3.35x C vs Fortran row vs col. ONF rewrite only, DNF fixed.

IV. PAPER V FIXES

Only OpenMP, OpenACC, OpenMPI deterministic.

V. INTERACTIVE GUI ARTIFACT

Open *moa_compiler/gui.html* in browser. Sliders d 4-36, k 16-256, n 128-4096 update live \$, bytes, times, energy, heat. Red dot current d . Green $M_{1exp} : 36$. For SC AE: zero-dep, check $d = 36, k = 64, n = 1024$ max proven screenshot, $d = 36, k = 256, n = 4096$ explosion, $d = 9, k = 64, n = 1024$ min optimal. See *MOA_GUI_USER_GUIDE.md*.

VI. CONCLUSION

Claim proven: Papers 1-5 went through MoA compiler in Python and Fortran (and C, OpenMP) with same DNF $F = dkn$ $M = dk$ $T = \alpha F + \beta M$ same output $F = 786432$ $M = 768$ $T = 0.006218\text{ms}$. Same DNF, many ONFs, predictable. Validated 5-device $M_{1exp}:36$ $Pd:9$ $Xd:0$ $Cd:36$. Think in MoA faster formal beautiful. Save human, machine, energy.

ACKNOWLEDGMENTS

NSF ACCESS CIS261342 (Delta A100 Sec VIII, MI100 Sec IX, Bridges-2 H100 Sec X, Expanse V100 Sec XI, Stampede3 Max1550 Sec XII) and CHI-261724 (Max1550) + UA A40.

Software: <https://github.com/womenflyplanes/moa-compiler> — *paper1_exact_reproduce.py*, *paper2_backward.py*, *paper3_decode.py*, *paper45_validation.py* —reference, *moa_sc_demo.ipynb*, *moa_compiler/gui.html*, *moa_figures/*,

generated/ 16 files Python Fortran C OpenMP same $F=786432$ $M=768$ $T=0.006218\text{ms}$ proof MULTI_LANG_PROOF.txt.

Claim: Papers 1-5 via compiler multi-lang Python and Fortran proven same DNF many ONFs predictable. 5-device $M_{1exp}:36$ $Pd:9$ $Xd:0$ $Cd:36$. See *run_all_langs.sh*. Think in MoA.

Fortran gfortran col-major gamma_col=7:

```
paper1_exact F=      786432  M=      768  T ms=
C row-major gamma_row=6:
paper1_exact F=786432 M=768 T=0.006218 ms bytes=3
OpenMP proc_bind(close) fixes 535x NUMA:
paper1_exact F=786432 M=768 T=0.006218 ms
DNF: ivec psi A = (rav A)[gamma(ivec, rhoA)] M1exp:
Artifact: moa_compiler/DELTA_REAL_HARDWARE_PROOF.t.
```

A. Delta Real Hardware Validation (A100 gpua027, A40 gpub017)

NCSA Delta validation on dt-login04 x86_64, gpua027 A100 (gpuA100x4), gpub017 A40 (gpuA40x4), account bihf-delta-gpu.

All 12 generated files identical $F = 786432$ $M = 768$ $T = 0.006218\text{ms}$, proving same DNF $\rightarrow$ many ONFs $\rightarrow$ predictable:

Fortran gfortran col-major gamma_col=7:

```
paper1_exact F=      786432  M=      768  T ms=
paper2_backward F=      786432  M=      768  T
paper3_decode F=      786432  M=      768  T m
paper45_matmul F=      786432  M=      768  T
C row-major gamma_row=6:
paper1_exact F=786432 M=768 T=0.006218 ms bytes=3
paper2_backward F=786432 M=768 T=0.006218 ms
paper3_decode F=786432 M=768 T=0.006218 ms
paper45_matmul F=786432 M=768 T=0.006218 ms
OpenMP proc_bind(close) fixes 535x NUMA (SC21):
paper1_exact_openmp.c F=786432 M=768 T=0.006218 m
paper2_backward_openmp.c F=786432 M=768 T=0.00621
paper3_decode_openmp.c F=786432 M=768 T=0.006218
paper45_matmul_openmp.c F=786432 M=768 T=0.006218
```

DNF: $\vec{\psi}A = (\text{rav } A)[\gamma(\vec{i}, \rho A)]$, $M_{1exp} : 36$
 $Pd : 9$ $Xd : 0$ $Cd : 36$, reached(d). Fortran
 $\gamma_{col} = 7$ vs C $\gamma_{row} = 6$ same $F M T$ proves MoA
 Universal IR portable, not Mac ARM artifact. Artifacts:
moa_compiler/DELTA_REAL_HARDWARE_PROOF.txt,
moa_A100_22646116.log,
moa_A40_22646132.log, *DELTA_DIRECT_RUN.log*.

Fortran gfortran col-major gamma_col=7:

```
paper1_exact F=      786432  M=      768  T ms=
C row-major gamma_row=6:
paper1_exact F=786432 M=768 T=0.006218 ms bytes=3
OpenMP proc_bind(close) fixes 535x NUMA:
paper1_exact F=786432 M=768 T=0.006218 ms
DNF: ivec psi A = (rav A)[gamma(ivec, rhoA)] M1exp:
Artifact: moa_compiler/DELTA_REAL_HARDWARE_PROOF.t.
```

B. Delta Real Hardware Validation (A100 gpua027, A40 gpub017)

NCSA Delta validation on dt-login04 x86_64, gpua027 A100 (gpuA100x4), gpub017 A40 (gpuA40x4), account bihf-delta-gpu.

All 12 generated files identical $F = 786432$ $M = 768$ $T = 0.006218\text{ms}$, proving same DNF $\rightarrow$ many ONFs $\rightarrow$ predictable:

```
Fortran gfortran col-major gamma_col=7:
paper1_exact F=      786432 M=      768 T ms=    6.21772837E-03
paper2_backward F=      786432 M=      768 T ms=    6.21772837E-03
paper3_decode F=      786432 M=      768 T ms=    6.21772837E-03
paper45_matmul F=      786432 M=      768 T ms=    6.21772837E-03
C row-major gamma_row=6:
paper1_exact F=786432 M=768 T=0.006218 ms bytes=3.145728 MB
paper2_backward F=786432 M=768 T=0.006218 ms
paper3_decode F=786432 M=768 T=0.006218 ms
paper45_matmul F=786432 M=768 T=0.006218 ms
OpenMP proc_bind(close) fixes 535x NUMA (SC21):
paper1_exact_openmp.c F=786432 M=768 T=0.006218 ms
paper2_backward_openmp.c F=786432 M=768 T=0.006218 ms
paper3_decode_openmp.c F=786432 M=768 T=0.006218 ms
paper45_matmul_openmp.c F=786432 M=768 T=0.006218 ms
```

DNF: $\vec{i}\psi A = (\text{rav } A)[\gamma(\vec{i}, \rho A)]$, $M1exp : 36$
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moa_A40_22646132.log, DELTA_DIRECT_RUN.log.